AND, $(p \to r) \wedge (q \to r)$ is True. ② r is False. By defn. of conditional, $(p \wedge q)$ is False. By defn. of AND, the following cases arises:

① Both p, q are False. By defn. of conditional, $(p \to r)$ is True and $(q \to r)$ is True. By defn. of AND, $(p \to r) \wedge (q \to r)$ is True.

② p is False, q is True. By defn. of conditional, $(p \to r)$ is True but $(q \to r)$ is False. By defn. of AND, $(p \to r) \wedge (q \to r)$ is False.

③ p is True, q is False. By defn. of conditional, $(p \to r)$ is False but $(q \to r)$ is True. By defn. of AND, $(p \to r) \wedge (q \to r)$ is False.

$\boxed{\therefore (p \wedge q) \to r \text{ and } (p \to r) \wedge (q \to r) \text{ are not logically equivalent}}$ □

37) Show that $(p \to q) \to (r \to s)$ and $(p \to r) \to (q \to s)$ are not logically equivalent.

Soln: Let $(p \to q) \to (r \to s)$ be False. By defn. of conditional, $(p \to q)$ is True and $(r \to s)$ is False. By defn. of conditional, r is True and s is False. By defn. of conditional, the following cases arise for the truth values of p and q: ① q is True. The truth value of p doesn't matter. Since r is True, by the defn. of conditional, $(p \to r)$ is True. Since s is False, by the defn. of conditional, $(q \to s)$ is False. By the defn. of conditional, $(p \to r) \to (q \to s)$ is False.

② q is False. By the defn. of conditional, p is False. By the defn. of conditional, $(q \to s)$ is True. By the defn. of conditional, $(p \to r) \to (q \to s)$ is True.

$\boxed{\therefore (p \to q) \to (r \to s) \text{ and } (p \to r) \to (q \to s) \text{ are not logically equivalent}}$ □

The **dual** of a compound proposition that contains only the logical operators $\vee$, $\wedge$ and $\neg$ is the compound proposition obtained by replacing each $\vee$ by $\wedge$, each $\wedge$ by $\vee$, each T by F, and each F by T. The dual of s is denoted by s^*.